

Study Guide: Introduction to Distributed Systems

This study guide provides a comprehensive review of the fundamental concepts, architectures, and challenges of distributed systems as outlined in the provided lecture materials. It includes a short-answer quiz, essay prompts, and a detailed glossary to assist in mastering the subject matter.

Part 1: Short-Answer Quiz

Instructions: Answer the following questions in 2-3 sentences based on the information provided in the source context.

1. **How is a distributed system defined in terms of its appearance and components?**
 2. **What are the two primary aspects of a distributed system's nodes?**
 3. **Why do nodes in a distributed system face synchronization and coordination problems?**
 4. **Compare the "Integrative" and "Expansive" views of building distributed systems.**
 5. **What is the technical distinction between a decentralized system and a distributed system?**
 6. **Explain the difference between a logically centralized system and a physically distributed one using the Domain Name System (DNS) as an example.**
 7. **What is an overlay network, and how do structured and unstructured types differ?**
 8. **Define "middleware" and its primary function within a distributed system.**
 9. **What are the three components of scalability that developers must consider?**
 10. **Why is it theoretically and practically impossible to achieve full distribution transparency regarding failures?**
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Part 2: Answer Key

1. **How is a distributed system defined in terms of its appearance and components?**
A distributed system is a piece of software consisting of a collection of autonomous computing elements. To the end user or application, these elements appear and operate as a single, coherent system.
2. **What are the two primary aspects of a distributed system's nodes?** The nodes are autonomous computing elements, which can be hardware devices or software processes. Because they are autonomous, they must collaborate to ensure the user perceives the collection as a single, unified system.
3. **Why do nodes in a distributed system face synchronization and coordination problems?** Nodes face these problems because each node is autonomous and possesses its own notion of time. There is no global clock in a distributed system, which leads to fundamental difficulties in managing time-sensitive actions across the group.
4. **Compare the "Integrative" and "Expansive" views of building distributed systems.**
The integrative view focuses on connecting existing, separate networked computer systems into a single, larger system. In contrast, the expansive view involves extending an existing networked computer system by adding additional computers to it.
5. **What is the technical distinction between a decentralized system and a distributed system?** In a decentralized system, processes and resources are *necessarily* spread

across multiple computers. In a distributed system, processes and resources are *sufficiently* spread until a required level of sufficiency is reached.

6. **Explain the difference between a logically centralized system and a physically distributed one using the Domain Name System (DNS) as an example.** A system can be logically centralized, meaning it follows a single ruling logic or order, while being physically distributed across many locations. For example, the roots of the DNS are logically centralized but physically massively distributed and decentralized across several organizations.
7. **What is an overlay network, and how do structured and unstructured types differ?** An overlay network is a collection of nodes where each node communicates only with its neighbors. In a structured overlay, each node has a well-defined set of neighbors (such as in a tree or ring), while in an unstructured overlay, nodes have references to randomly selected other nodes.
8. **Define "middleware" and its primary function within a distributed system.** Middleware is described as the "Operating System of Distributed Systems," sitting as a layer between applications and local operating systems. Its primary function is to provide commonly used components and functions so that applications do not have to implement them separately.
9. **What are the three components of scalability that developers must consider?** Developers must account for size scalability (the number of users or processes), geographical scalability (the maximum distance between nodes), and administrative scalability (the number of administrative domains involved).
10. **Why is it theoretically and practically impossible to achieve full distribution transparency regarding failures?** Full transparency is impossible because a system cannot reliably distinguish between a very slow computer and one that has failed. Furthermore, it is never certain if a server successfully performed an operation immediately before a crash occurs.

Part 3: Essay Questions

Instructions: Use the concepts discussed in the source material to provide detailed responses to the following prompts.

1. **The Transparency-Performance Trade-off:** Discuss why aiming for full distribution transparency can negatively impact a system's performance. Consider the roles of communication latency, replica synchronization, and fault tolerance in your answer.
2. **Challenges of Geographical Scalability:** Analyze the specific hurdles faced when transitioning a distributed system from a Local Area Network (LAN) to a Wide Area Network (WAN), specifically regarding communication reliability and interaction schemes.
3. **The Role of Openness in Distributed Systems:** Explain what it means for a system to be "open" and why achieving independence from the heterogeneity of hardware, platforms, and languages is difficult for developers.
4. **Centralized Solutions—Misconceptions and Reality:** Critique the common misconceptions that centralized solutions do not scale and are always single points of failure. Provide evidence for how these systems can be made robust.

5. **Scaling Techniques—Latency and Consistency:** Evaluate the techniques used to improve size scalability, such as asynchronous communication and replication. Discuss the "consistency problem" that arises when multiple copies of data exist.

Part 4: Glossary of Key Terms

Term	Definition
Administrative Scalability	The ability of a distributed system to span across multiple independent administrative domains with conflicting policies.
Asynchronous Communication	A communication method where the sender does not wait for a response before continuing with other tasks, used to hide latencies.

**Autonomous
Computing
Elements**

Also known as nodes; hardware or software units that operate independently within a distributed system.

Coherent System

A system where a collection of nodes operates the same regardless of where, when, or how the interaction takes place.

**Distribution
Transparency**

The goal of hiding the distributed nature of a system (such as node location or failure) from users and applications.

Geographical Scalability	The ability of a system to maintain performance as the physical distance between its nodes increases.
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Middleware	A software layer providing common services and interfaces to applications, functioning as an intermediary between the OS and the app.
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Node	A single hardware device or software process that acts as an autonomous element within a distributed system.
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Open Distributed System

A system that can interact with services from other systems by conforming to well-defined interfaces and supporting portability.

Overlay Network

A virtual network where nodes communicate only with a specific set of neighbors within the larger collection.

Partial Failure

A situation where only a portion of a distributed system fails; hiding these failures is a major challenge in maintaining coherence.

Replication	The process of making copies of data or services available on different machines to increase availability and scalability.
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Size Scalability	The capacity of a system to handle an increasing number of users, processes, or resources.
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Structured Overlay	An overlay network where neighbors are assigned based on a well-defined structure, such as a ring or a tree.
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**Unstructured
Overlay**

An overlay network where each node maintains references to other nodes selected at random.

NotebookLM can be inaccurate; please double check its responses.